Exact Structures and Maximal Canonically Jordan Recoverable Subcategories for Modules over Type A Algebras

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Based on joint work with Benjamin Dequêne
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Plan

I. Motivation: Jordan Recoverability and Canonical Jordan Recoverability

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II. The Diamond Exact Structure $\mathcal{E}_\diamond$

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(First Main Theorem)

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In recent work, **Garver–Patrias–Thomas** (2018–2022) introduced and developed the notions of *Jordan recoverability (JR)* and *canonical Jordan recoverability (CJR)*. These ideas were later expanded by **Dequêne** (2022–2024).

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- $\mathscr{A}$ will denote a **hereditary abelian category**.
- All subcategories are assumed **full, closed under finite direct sums, and under direct summands.**

Jordan Form for Representations

Let $E = (E_q, E_\alpha) \in \mathrm{rep}Q$ *and* $N = (N_q)_{q \in Q_0} \in \mathrm{NEnd}(E)$*.* Each N_q is a nilpotent endomorphism of E_q.

Definition

The **Jordan form** of N_q is the partition $JF(N_q) \vdash \dim(E_q)$ recording its Jordan block sizes. Define the tuple

$$JF(N) \;=\; (JF(N_q))_{q \in Q_0},$$

which describes the block structure of N at each vertex.

The Generic Jordan Form Invariant

Fix $E \in \mathrm{rep} Q$. *For* $N = (N_q)_{q \in Q_0} \in \mathrm{NEnd}(E)$, *set*

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Definition [GPT23]

The ***generic Jordan form*** *of E is the maximal value (with respect to dominance order):*

$$\mathrm{GenJF}(E) = \max_{N \in \mathrm{NEnd}(E)} JF(N).$$

Question : Is this invariant complete?

Jordan Recoverability

Definition [GPT23]

Let $\mathscr{C} \subseteq \mathrm{rep}Q$ *be a subcategory.* We say that $\mathscr{C}$ is **Jordan recoverable (JR)** if

$$E \simeq F \iff \mathrm{GenJF}(E) = \mathrm{GenJF}(F) \qquad \text{for all } E, F \in \mathscr{C}.$$

Remark

The generic Jordan form GenJF is a *complete invariant* on $\mathscr{C}$.

Canonical Jordan Recoverability (CJR)

Definition [GPT23]

Let $\mathscr{C} \subseteq \mathrm{rep}Q$ We say $\mathscr{C}$ is ***canonically Jordan recoverable (CJR)*** *if for every $X \in \mathscr{C}$ there exists a dense open set (in the Zariski sense) $\Omega \subseteq \mathrm{rep}(Q, \mathrm{GenJF}(X))$ such that for all $Y \in \Omega$, $Y \simeq X$.*

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Maximal CJR subcategory

*A CJR subcategory $\mathscr{C}$ is **maximal** if it is not properly contained in any larger CJR subcategory of $\mathrm{rep}Q$, i.e.*

$$\forall \mathcal{D} \supsetneq \mathscr{C}, \quad \mathcal{D} \text{ is not CJR.}$$

Example on A_2: JR but not CJR

Assume $Q : 1 \to 2$ and let $\mathscr{C} = \mathrm{add}(S_1, S_2) \subseteq \mathrm{rep}(Q)$. Every $E \in \mathscr{C}$ has the form

$$E = K^a \xrightarrow{0} K^b.$$

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Generic Jordan form

$$\mathrm{GenJF}(E) = \big((a), (b)\big).$$

Hence $\mathscr{C}$ is **JR**: $E \simeq F \iff \mathrm{GenJF}(E) = \mathrm{GenJF}(F)$ *in* $\mathscr{C}$*.*

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Open sets with generic Jordan form $(1, 1)$.

Consider $S_1 \oplus S_2 \cong K \xrightarrow{0} K$. Any dense open $U \subset \mathrm{rep}_K(Q, (1, 1))$ contains representations $K \xrightarrow{\lambda} K \cong P_1$ with $\lambda \neq 0$.

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We have $\mathrm{GenJF}(S_1 \oplus S_2) = ((1), (1)) = \mathrm{GenJF}(P_1)$ with $P_1 \in U$, but $S_1 \oplus S_2 \not\cong P_1$. Hence $\mathscr{C}$ is **not CJR**.

Maximal CJR subcategories in type A_n

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Theorem [Deq23a]

A subcategory $\mathscr{C} \subseteq \mathrm{rep}(Q)$ is canonically Jordan recoverable if and only if every non-split short exact sequence

$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0$$

with $X, Z \in \mathscr{C}$, $Y \notin \mathrm{ind}(\mathrm{rep}\, Q)$.

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$$\begin{tikzcd} 0 \arrow[r] & X \arrow[r, tail, "f"] \arrow[d, "h"'] & Y \arrow[r, two heads] \arrow[d] & Z \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & D \arrow[r, tail] & PO \arrow[r, two heads] & Z \arrow[r] & 0 \end{tikzcd}$$

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(ES3-II) Pullbacks

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Example of Exact Structures

$\mathcal{E}_{\max}$: the *maximal exact structure*, containing all short exact sequences.

$\mathcal{E}_{\min}$: the *minimal exact structure*, containing only the split short exact sequences.

Short Exact Sequences in Type A_n

Theorem

Let Q be an A_n type quiver. For a non-split exact sequence

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In case (b) we call it a **diamond exact sequence**.

The Diamond Exact Structure $\mathcal{E}_\diamond$

Definition

Let Q be an A_n type quiver. We define the collection of short exact sequences $\mathcal{E}_\diamond$ given by the additive bifunctor $\mathcal{E}_\diamond(-,-)$, uniquely determined by

$$\mathcal{E}_\diamond(N,M) = \{\, \eta \in \mathrm{Ext}^1(N,M) \mid \eta \text{ splits or diamond exact sequence} \,\}$$

for $M, N \in \mathrm{ind}(\mathrm{rep}\, Q)$ indecomposable.

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Proposition [Brüstle–Hanson–R.–Schiffler, 2024]

Let $n \in \mathbb{N}^*$ and Q be an A_n type quiver. Then $\mathcal{E}_\diamond$ is an exact structure on $\mathscr{A} = \operatorname{rep}(Q)$.

Restatement of Theorem on CJR subcategories

Theorem

Let Q be an A_n type quiver. A subcategory $\mathscr{C} \subseteq \mathrm{rep}(Q)$ is canonically Jordan recoverable if and only if $\mathscr{C}$ is $\mathcal{E}_\diamond$-adapted.

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Remark. $\mathscr{C}$ is maximal canonically Jordan recoverable if any subcategory $\mathscr{D} \supset \mathscr{C}$ is not $\mathcal{E}_\diamond$-adapted.

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For an object $M \in \mathscr{A}$, we set $\mathrm{Gen}_{\mathcal{E}}(M) = \mathrm{Gen}_{\mathcal{E}}(\mathrm{add}(M))$, $\mathrm{Sub}_{\mathcal{E}}(M) = \mathrm{Sub}_{\mathcal{E}}(\mathrm{add}(M))$.

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(b) For $i \geq 1$, define

$$\mathrm{GS}^i_{\mathcal{E}}(\mathscr{C}) = \mathrm{add}\Big(\mathrm{Gen}_{\mathcal{E}}(\mathrm{GS}^{i-1}_{\mathcal{E}}(\mathscr{C})) \,\cup\, \mathrm{Sub}_{\mathcal{E}}(\mathrm{GS}^{i-1}_{\mathcal{E}}(\mathscr{C}))\Big).$$

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Remark. *$(\mathrm{GS}^i_{\mathcal{E}}(\mathscr{C}))_{i\in\mathbb{N}}$ is an increasing sequence of subcategories of $\mathscr{A}$.*

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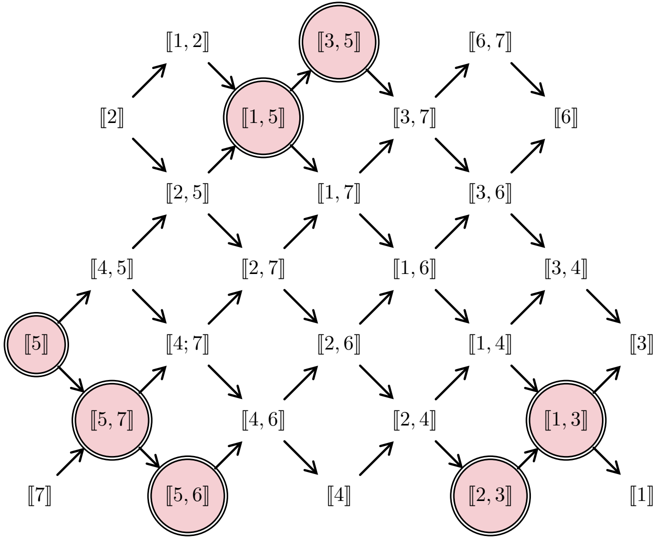
For an object $M \in \mathscr{A}$, we set $\mathrm{GS}_{\mathcal{E}}(M) = \mathrm{GS}_{\mathcal{E}}(\mathrm{add}(M))$.

$\mathrm{GS}^0_{\mathcal{E}_\diamond}(\circledcirc) = \mathrm{add}(\circledcirc)$

$$Q = 1 \rightarrow 2 \leftarrow 3 \rightarrow 4 \rightarrow 5 \leftarrow 6 \rightarrow 7$$



$\mathrm{GS}^1_{\mathcal{E}_\diamond}(\bigcirc)$

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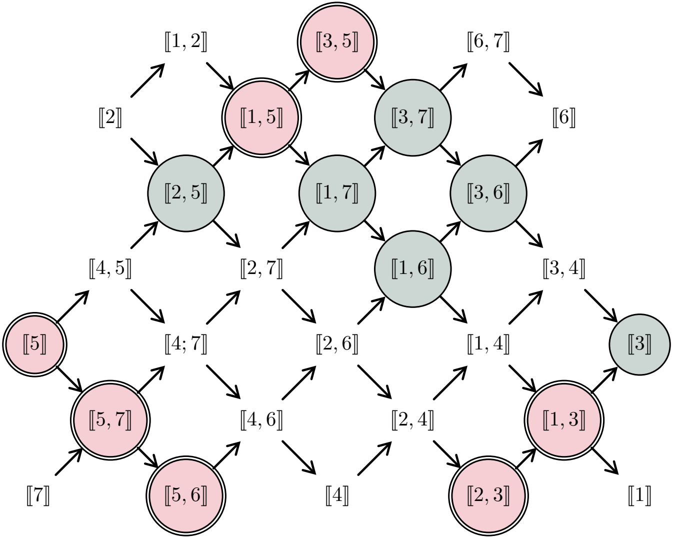

$\mathrm{GS}_{\mathcal{E}_\diamond}(\bigcirc) = \mathrm{GS}^2_{\mathcal{E}_\diamond}(\bigcirc)$

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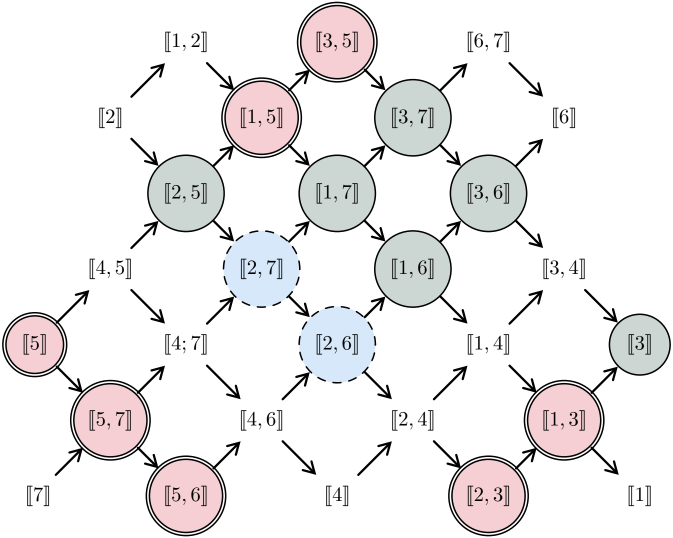

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(1) If $\mathscr{C}$ is $\mathcal{E}$-adapted, then $\mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ is also $\mathcal{E}$-adapted.

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Corollary 1. *Let $T \in \mathscr{A}$ be a tilting object. Then $\mathrm{GS}_{\mathcal{E}}(T)$ is $\mathcal{E}$-adapted and extension-closed.*

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Corollary 1. *Let $T \in \mathscr{A}$ be a tilting object. Then $\mathrm{GS}_{\mathcal{E}}(T)$ is $\mathcal{E}$-adapted and extension-closed.*

Corollary 2. *Let $T \in$ rep(A_n) be a tilting object. Then $\mathrm{GS}_{\mathcal{E}_\diamond}(T)$ is canonically Jordan recoverable.*

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Let $\mathscr{C} \subseteq \mathscr{A}$ be a subcategory.

(1) If $\mathscr{C}$ is $\mathcal{E}$-adapted, then $\mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ is also $\mathcal{E}$-adapted.

(2) If $\mathscr{C}$ is $\mathcal{E}$-adapted and extension-closed, then $\mathrm{GS}_{\mathcal{E}}(\mathscr{C})$ is also extension-closed.

Corollary 1. *Let $T \in \mathscr{A}$ be a tilting object. Then $\mathrm{GS}_{\mathcal{E}}(T)$ is $\mathcal{E}$-adapted and extension-closed.*

Corollary 2. *Let $T \in$ rep(A_n) be a tilting object. Then $\mathrm{GS}_{\mathcal{E}_\diamond}(T)$ is canonically Jordan recoverable.*

Question: Is $\mathrm{GS}_{\mathcal{E}_\diamond}(T)$ maximal canonically Jordan recoverable?

First Main Theorem

Theorem (Dequêne–R, 2025)

Let $T \in \mathscr{A}$ be a tilting object. Then $\mathrm{GS}_{\mathcal{E}}(T)$ is a maximal $\mathcal{E}$-adapted, extension-closed subcategory of $\mathscr{A}$ containing T.

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Corollary

For any tilting object $T \in \mathscr{A}$,

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How can we describe the equivalence relation :

$$T \sim_{\mathcal{E}} T' \iff \mathrm{GS}_{\mathcal{E}}(T) = \mathrm{GS}_{\mathcal{E}}(T')?$$

Plan

I. Motivation: Jordan Recoverability and Canonical Jordan Recoverability

II. The Diamond Exact Structure $\mathcal{E}_\diamond$

III. The $\mathrm{GS}_\mathcal{E}$ Operator & Properties
(First Main Theorem)

IV. Relative Mutations & Tilting Equivalence
(Second Main Theorem)

Classical and Relative Mutation

Definition

Let $T = U \oplus \overline{T}$ be a tilting object with U indecomposable.

*A **mutation** of T at U is a tilting object*

$$\mu_U(T) \;=\; U' \oplus \overline{T},$$

where U' is an indecomposable object obtained from U by an exchange sequence

$$\xi :\; 0 \to U \xrightarrow{f} B \xrightarrow{g} U' \to 0\,, \qquad B \in \mathrm{add}(\overline{T}),$$

where

f is a left minimal $\mathrm{add}(\overline{T})$-approximation of U,

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We write $\mu_U^+(T)$ (resp. $\mu_U^-(T)$) for the *right* (resp. *left*) mutation.

Classical and Relative Mutation

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*An $\mathcal{E}$-**mutation** of T at U is a tilting object*

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where U' is an indecomposable object obtained from U by an $\mathcal{E}$-exchange sequence

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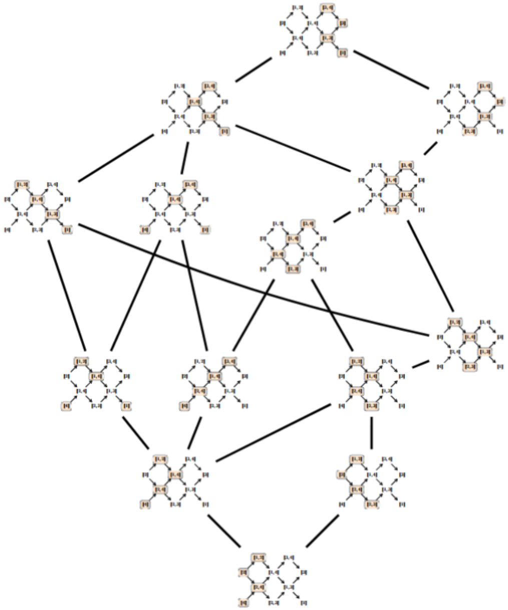
We write $\mu^{+}_{U,\mathcal{E}}(T)$ (resp. $\mu^{-}_{U,\mathcal{E}}(T)$) for the *right* (resp. *left*) $\mathcal{E}$-mutation.

Example : $Q = 1 \to 2 \leftarrow 3 \to 4$ with $\mathcal{E} = \mathcal{E}_{\max}$



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T' *is* $\mathcal{E}$-***reachable*** *from* T *if there exist indecomposables* $(U_1, \ldots, U_m)$ *such that*

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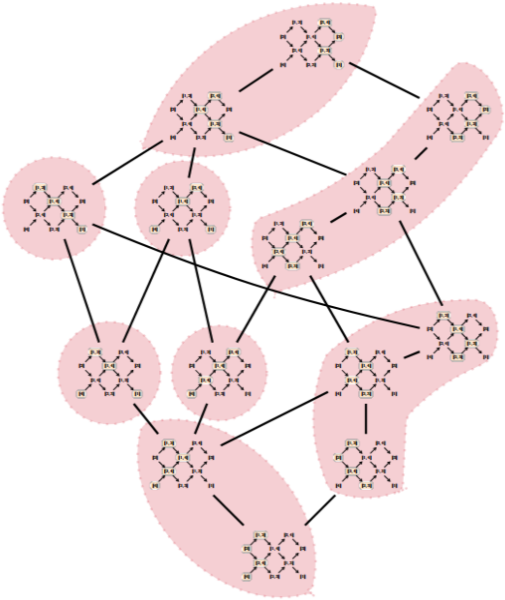
$\mathcal{E}$-**reachability** : $T \approx_{\mathcal{E}} T'$ forms a equivalence relation. We denote $[T]_{\approx_{\mathcal{E}}}$ for the equivalence class of T.

Example : $Q = 1 \to 2 \leftarrow 3 \to 4$ with $\mathcal{E} = \mathcal{E}_\diamond$



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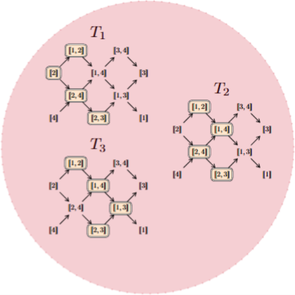

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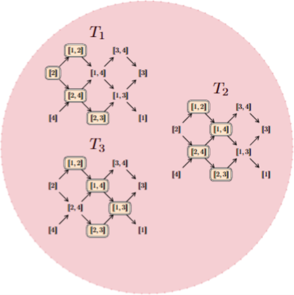

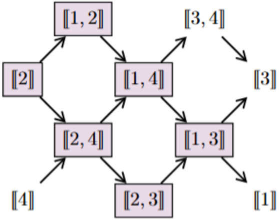

Second Main Theorem

Theorem (Dequêne–R, 2025)

Let $T \in \mathrm{Tilt}(Q)$. *Then*

$$\mathrm{GS}_{\mathcal{E}}(T) \;=\; \mathrm{add}\left(\bigoplus_{T' \in [T]_{\approx_{\mathcal{E}}}} T' \right).$$

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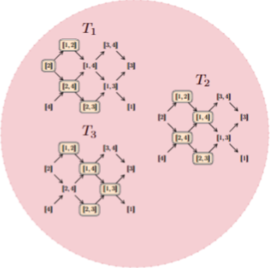
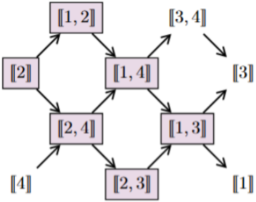


$\mathrm{add}(T_1 \oplus T_2 \oplus T_3)$



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(b) Let $\mathscr{C} \subseteq \operatorname{rep}(Q)$ be a maximal canonically Jordan recoverable subcategory. Then there exists $T \in \operatorname{Tilt}(Q)$ such that

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Questions?

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